

Comprehensive Sensitivity Analysis Report

Week 7 evaluates how the JPM chooser model and revised machine-learning routes respond to volatility, interest rates and VIX. The required shocks increase the structural BSM price, while the ML routes respond differently. The analysis also records complete-feature update and fallback controls.

Reference Contract and Observation

Input	Value
Observation	2024-11-29, last complete Week 5 target row
JPM spot and strike	\$241.1194 and \$150
Choice and maturity	T1 = 0.5 year; T2 = 1 year
Volatility and rate	45.7249% annual volatility; 4.18% Treasury proxy
Dividend yield	Paper assumption $q = 2.33\%$

Structural BSM Shocks

The volatility shock multiplies annualized volatility by 1.5. The rate shock adds 0.02 to the decimal rate, taking 4.18% to 6.18%; it is an absolute two percentage-point rise. Spot, strike, dividend yield and contract dates remain fixed. The combined scenario applies both changes.

Scenario	BSM price USD	Change USD	Change percent
Base	99.2617	+0.0000	+0.00%
50% Volatility Spike	115.6683	+16.4066	+16.53%
2% Rate Hike	101.3317	+2.0700	+2.09%
Combined Shock	116.9177	+17.6561	+17.79%

The volatility shock increases the chooser price by \$16.4066, compared with \$2.0700 for the rate shock. These are finite scenario changes at the stated contract, not universal sensitivities or observed trading profits. The combined change is not assumed to equal the sum of separate shocks.

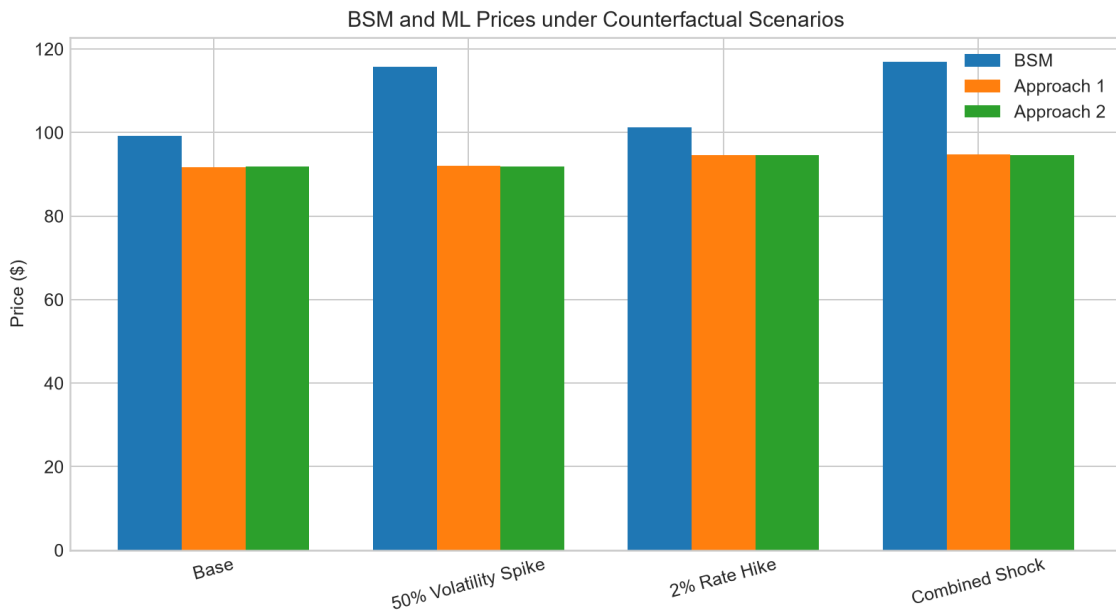
Machine Learning Counterfactuals

Approach 1 retains the original Random Forest volatility forecast plus BSM. Approach 2 now learns bounded normalized time value directly, with regularization and controlled moneyness tails. This analysis reruns the repaired Week 6 model; no dividend-growth feature or upstream data replacement is introduced.

For the ML volatility scenario, all rolling-volatility features and VIX level are multiplied by 1.5, with VIX return set to at least 0.50. The rate scenario adds 0.02 to the decimal Treasury feature and 2.0 to its percentage-point momentum. Other features stay fixed. These coordinated feature shocks differ from a one-input BSM sensitivity curve.

Scenario	BSM USD	RF plus BSM USD	Direct price USD
Base	99.2617	91.7072	91.8375
50% Volatility Spike	115.6683	92.0449	91.9144
2% Rate Hike	101.3317	94.5557	94.5690
Combined Shock	116.9177	94.8010	94.5776

Figure 1 Revised model counterfactual prices



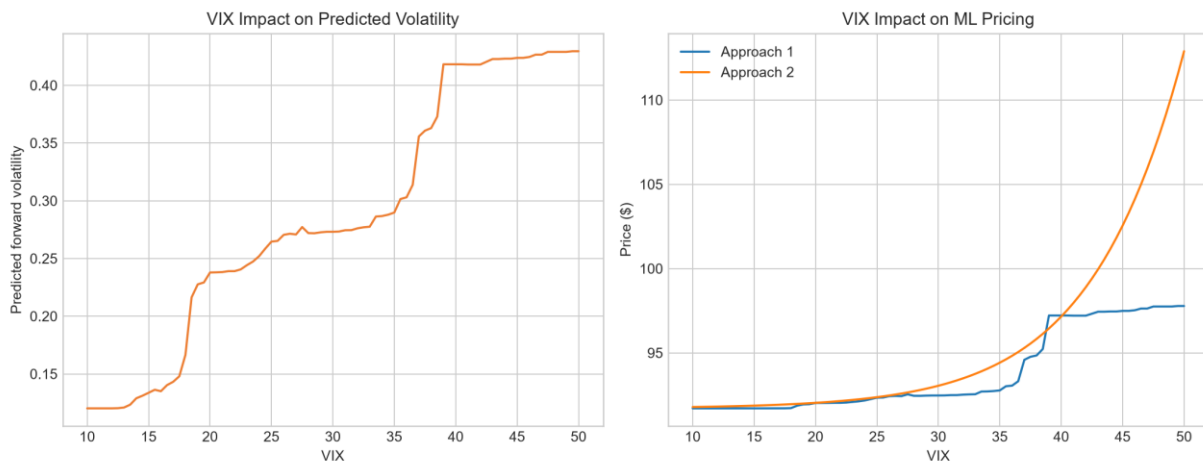
The RF route rises by \$0.3378 under the volatility shock. The revised direct route predicts within structural price bounds, without clipping negative output to zero. These counterfactual prices reflect different fitted mechanisms and are not observed market responses.

VIX Sensitivity and Model Interpretation

Explained model	VIX share of mean absolute SHAP
Approach 1 Random Forest	40.80%
Approach 2 bounded regression	23.29%

VIX-related SHAP shares are 40.80% for the RF volatility model and 23.29% for the complete bounded direct-price predictor. Direct SHAP uses a training-only background and permutation approximation. The targets and units differ across routes; compare within-model shares, not raw SHAP magnitudes or causal effects.

Figure 2 VIX level counterfactual on the reference observation



The saved VIX grid spans 10.0 to 50.0. Only VIX level varies in this grid; other features retain their reference values. Approach 1 prices range from \$91.7071 to \$97.7777. Tree thresholds can produce flat segments and jumps, so this curve should not be read as a smooth Greek.

Counterfactual rows can lie outside observed feature combinations. The fixed \$150 strike is also far below the reference spot. These constraints matter when interpreting both weak price responses and extrapolation. VIX remains a volatility indicator; the unavailable news-sentiment series is not synthesized or substituted.

Complete Feature Updates and Failure Controls

Both tools retain the original 17 volatility and 19 direct-price input columns. Histories rebuild all features at a common completed date. The direct estimator transforms moneyness internally and separates the online pricing rate from the DGS10 feature. The research sample and Approach 1 are unchanged; Approach 2 is explicitly revised.

Control	Implementation
Request bounds	12-second timeout and at most two attempts per request
Source independence	Parallel requests and a separate atomic JSON cache per source
Missing source	Use its last available history; label cached or unavailable status
Incomplete feature row	Retain the prior complete snapshot rather than combine current spot with old features
Rate meaning	DGS10 stays the trained ML feature; DGS1 supplies the one-year BSM pricing rate
Automatic refresh	On opening and every 15 minutes during an active app session; manual refresh remains available
Credentials	Read FRED key from environment or local .env; omit keys and authenticated URLs from logs

On 10 September 2026, both updater paths returned all four sources successfully and produced a complete feature row dated 8 September 2026, the latest common date. Tests also covered source timeout, per-source fallback, corrupt cache and historical feature parity. Provider delays can still occur; a successful update is not a guarantee of intraday freshness.

Conclusion and Evidence

The required stresses, regenerated ML counterfactuals, VIX analysis and complete update controls are delivered. News sentiment and actual chooser transactions remain limitations. Scheduled refresh is configured; this model revision requires publication after the report PDFs are regenerated.

Evidence: Week7_Advanced_Analysis.ipynb; sensitivity_analysis.py;
model_results/structural_stress_scenarios.csv; ml_counterfactual_scenarios.csv;
shap_feature_group_impact.csv; vix_counterfactual_curve.csv; .github/scripts/test_feature_refresh.py. All prices are model estimates in USD and are not trading recommendations.